

EDUCATION

Stanford University

Master of Science, Mechanics and Computation

Autumn 2025 to Winter 2027

Stanford, CA, USA

• **Relevant Coursework:** Solid Mechanics, Linear Algebra with applications to Engineering Computations, Advanced Structural Analysis

Birla Institute of Technology & Science, Pilani (BITS Pilani)

Bachelor of Engineering, Civil Engineering — **CGPA: 9.26/10**

Aug 2021 - May 2025

Pilani, India

• **Relevant Coursework:** Mechanics of Solids, Analysis of Structures, Design of Reinforced Concrete Structures, Foundation Engineering, Design of Steel Structures, Earthquake Resistant Design and Construction, Introduction to FEM

TECHNICAL SKILLS

Software/ Tools: DIANA FEA, ABAQUS, ETABS, MASTAN2, AutoCAD, Revit, MS Office Suite, Git, Github

Programming: MATLAB, Python, Julia, C, $\LaTeX$

PUBLICATIONS

- *Experimental and numerical studies on reinforced concrete base restrained walls and tanks subjected to shrinkage effects* Amrit Bahl, Mohammad Najeeb Shariff, and Sankati Yellamanda
Accepted in the *ACI Structural Journal*, Preprint available: International Concrete Abstracts Portal
- *A comparison of analytical models for crack-width prediction in RC ties subject to direct tension and restrained shrinkage* A R Bahl, Pawan Prakash, and M N Shariff
Accepted in the *4th R.N. Raikar Memorial International Conference 2024*
- *Dynamic stability of randomly distributed CNT reinforced composite beam using the generalized differential quadrature method* A R Bahl, and Rajesh Kumar
Accepted as a book chapter in the book *Experimental and Mathematical Modeling of Aerospace and Civil Engineering Composite Structures* by Apple Academic Press (CRC Press/ Taylor and Francis)

RESEARCH EXPERIENCE

Design optimization of composite compressed air energy storage (CAES) pressure vessel Jun 2025 - Aug 2025

Research Assistant: Mentor: Dr. Rajesh Kumar, BITS Pilani — Sponsored by: Siemens Energy

- Validated the burst pressure and stress-strain behavior of CFRP and GFRP-Steel composite pressure vessels with experimental results using a progressive failure based non-linear finite element analysis in ABAQUS
- Conducted a parametric study on the designed prototype with joints and holes to find the optimum fiber winding angle that reduced the total material used to sustain a required design pressure of 110 bar

Numerical study on RC base-restrained walls and tanks subjected to shrinkage effects Jun 2024 - Dec 2024

Undergraduate Thesis: Mentor: Dr. Mohammad Najeeb Shariff, Indian Institute of Technology Bombay

- Compared existing analytical models in direct tension and restrained shrinkage to determine the accuracy and effect of the various parameters on crack-width using Sobol's sensitivity analysis
- Validated the crack-widths, crack-patterns, and shrinkage strains from the experimental tests done on base-restrained walls and tanks using a 3D nonlinear finite element (FE) analysis on DIANA FEA
- Fitted an equation to find the variation of the restraint factor along the height of a wall for different aspect ratios derived using a linear FE analysis to assess the tensile stresses due to a base-restraint

Dynamic stability analysis of randomly distributed CNT reinforced composite beam Jan 2024 - May 2024

Research Assistant: Mentor: Dr. Rajesh Kumar, BITS Pilani

- Modeled the effective three-phase CNT-Fiber-Composite system using the Eschelby-Mori-Tanaka scheme and the Chamis homogenization technique
- Solved the governing differential equations for the First-Order Shear Deformation theory using the Generalized Differential Quadrature method to validate the natural frequency and critical buckling load
- Conducted a parametric study to find the effect of agglomeration ratio on the instability regions using Bolotin's method

ACADEMIC ACHIEVEMENTS

Poster Presentation Competition - Young Researchers Conclave

- Awarded the second position in the poster presentation competition organized as a part of the Young Researchers Conclave at the CSIR-CRRI for the poster titled *Stabilization of Black Cotton Soil with Alkali Bypass dust*